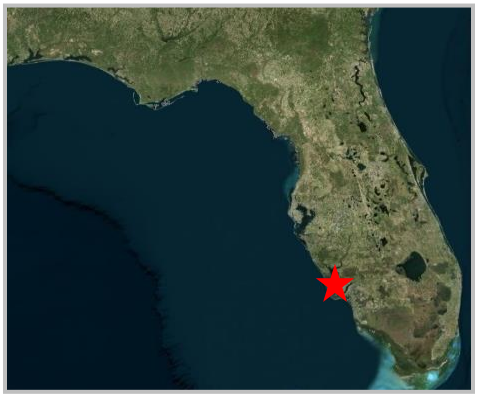
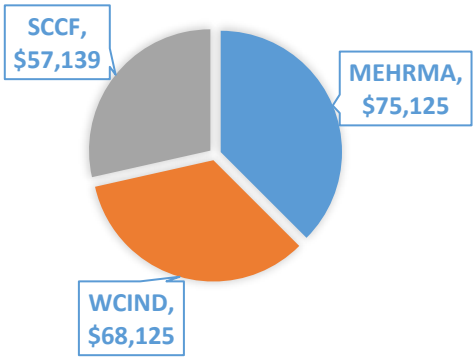
Oyster Reef Restoration in Southwest Florida, San Carlos Bay, Lee County, FL

FY 17/18



Project Partners and Funding



Project Description

San Carlos Bay is a 10,000-acre waterbody between the mouth of the Caloosahatchee River, Sanibel Island, Pine Island, and the Gulf of Mexico. The project area included an oyster reef and adjacent submerged lands. The purpose of this project was to restore a large degraded oyster reef. This project built upon the success of two prior oyster reef restoration projects conducted by the Sanibel-Captiva Conservation Foundation (SCCF) in both San Carlos Bay and nearby Tarpon Bay. This project employed the same restoration techniques used in the previous successful restorations. Fossilized oyster shell (approximately 700 yd3 of 4'' shell) was added to the existing degraded reef to increase reef elevation and provide oyster colonization substrate over a one acre area.



Figure 1: Volunteers with Sanibel-Captiva Conservation Foundation placed recycled oyster shell on reefs. December 14, 2017.

Additional project funding was contributed by the West Coast Inland Navigation District through the Lee County Natural Resources Department. To increase awareness of the importance of oyster reefs, SCCF held four volunteer oyster reef restoration events. Monitoring will be conducted by SCCF for 2 years following the guidelines in the Oyster Habitat Restoration Monitoring and Assessment Handbook (Baggett et al. 2014).

Benefits

- Attenuate waves and boat wakes in shallow seagrass habitat
- Reduce turbidity over seagrass in the project area
- Provide substrate for the recruitment and colonization of oysters, sponges, and algae

- Enhance nursery and foraging habitat for a variety of wildlife including juvenile fish, invertebrates, oysters, wading birds, manatees and juvenile green sea turtles.

FWC Strategic Plan Elements Supported

Theme 1, Goal 1. Ensure the sustainability of Florida's fish and wildlife populations.

Strategy:

- Manage non-listed species to prevent them from meeting Florida's listing criteria for endangered and threatened species.

Theme 1, Goal 2. Ensure sufficient habitats exist to support healthy and diverse fish and wildlife populations.

Strategy:

- Manage habitats to sustain healthy and diverse fish and wildlife populations.

Theme 3, Goal 3. Increase opportunities for residents and visitors, especially youth, to actively support and practice fish and wildlife conservation stewardship.

Strategy:

- Provide and promote opportunities for residents and visitors, especially youth, to participate in conservation stewardship activities, including volunteer opportunities with the FWC.

Theme 3, Goal 4. Encourage communities to conserve lands and waters critical to sustaining healthy and diverse populations of fish and wildlife.

Strategies:

- Support community events and programs that promote fish and wildlife conservation.

Acreage restored by habitat type (footprint)

Total- 0.93 acres oyster reef:

- Oyster habitat- 0.93 acre

Acreage affected by restoration activities: 6.74 acres



Figure 2: Newly constructed reef used by American oystercatchers. March 19, 2018.

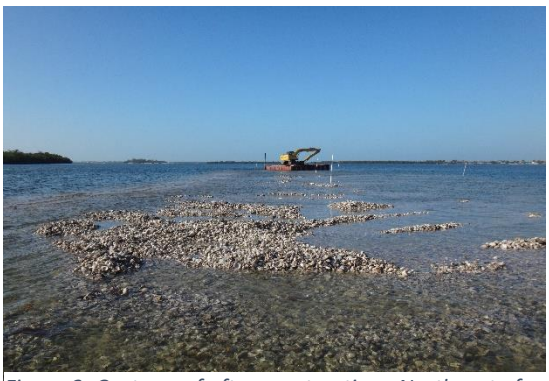


Figure 2: Oyster reef after construction. Northeast of project site viewed to the southwest. April 2, 2018.